Universal Consonance in Exoplanet Orbital Architectures: Evidence for Farey Hierarchy Locking Across 1,038 Multi-Planet Systems

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Abstract

We present a systematic analysis of orbital period ratios across 1,038 confirmed multi-planet systems from the NASA Exoplanet Archive, comprising 2,593 planets and 2,344 unique pairwise period ratios. We test whether these ratios preferentially cluster near simple integer fractions (consonant ratios p/q with q <= 12), as predicted by a universal consonance hierarchy in which locking strength scales inversely with consonance rank (p + q). We find that 1,967 of 2,344 ratios (83.9%) fall within 2% of a consonant fraction. Against a log-uniform null distribution – the physically appropriate comparison for periods spanning orders of magnitude – this yields z = 43.08 (p < 10^{-4}, 10,000 Monte Carlo trials). The five most populated ratios are 1:2 (140 hits), 1:3 (108), 2:3 (96), 1:4 (83), and 1:5 (82), precisely the lowest-rank fractions in the Farey hierarchy. Known resonant chains (TRAPPIST-1, Kepler-223, HD 110067, TOI-178, GJ 876, Kepler-80) are recovered with deviations < 0.1%. These results constitute the first population-level confirmation that exoplanet orbital architectures follow a universal consonance hierarchy, extending mean-motion resonance from a property of select systems to a statistical attractor across the full exoplanet census.

Keywords: exoplanets, orbital resonance, mean-motion resonance, Farey sequence, period ratios, planetary dynamics

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1. Introduction

1.1 Mean-Motion Resonance in Planetary Systems

Mean-motion resonances (MMRs) – configurations in which the orbital periods of two planets form a ratio of small integers – have been recognized as dynamically significant since the study of Jupiter's Galilean moons. In the exoplanet era, MMRs have been identified in remarkable multi-planet chains: the seven-planet TRAPPIST-1 system exhibits a continuous chain of near-integer period ratios [1], Kepler-223 hosts a four-planet 3:4:6:8 chain [2], and the recently characterized HD 110067 system displays six planets locked in a pristine resonant chain [3]. TOI-178 [4], GJ 876 [5], and Kepler-80 [6] provide additional well-studied examples.

The theoretical basis for MMR is well established. Convergent migration in protoplanetary disks drives planet pairs toward commensurabilities, where resonant perturbations can capture and maintain the configuration [7,8]. The strength of resonant locking depends on the order of the resonance: first-order resonances (period ratios of the form (k+1):k) are the strongest attractors, with higher-order resonances progressively weaker [9].

1.2 Beyond Individual Chains: A Universal Hierarchy?

Despite extensive study of individual resonant systems, a systematic question remains largely unaddressed: does the full population of multi-planet systems exhibit a statistically significant preference for consonant period ratios, and if so, does this preference follow a predictable hierarchy?

Previous population-level studies have noted that Kepler multi-planet systems show period ratios clustered near, but slightly wide of, first-order MMRs [10,11]. Fabrycky et al. [12] documented the period ratio distribution of Kepler multis, finding peaks near 2:1 and 3:2 with characteristic asymmetries attributed to resonance repulsion. However, these analyses typically restrict attention to first- and second-order resonances and do not test the full spectrum of integer ratios against a quantitative hierarchy.

We propose a more general framework: the **Consonance Principle**, which predicts that any two gravitationally coupled periodic orbits will tend toward rational period ratios p:q, with the locking strength inversely proportional to the **consonance rank** r_c = p + q. This hierarchy is identical to the Farey sequence ordering of rational numbers and arises generically in coupled oscillator systems [13,14]. The prediction is not merely that resonances exist, but that they populate a specific hierarchy: lower-rank ratios (1:2, 1:3, 2:3) should be more strongly attractive than higher-rank ratios (5:11, 7:9), producing a monotonically decreasing frequency-of-occurrence with increasing rank.

1.3 This Work

We test the Consonance Principle against the complete NASA Exoplanet Archive catalog of confirmed multi-planet systems. We compute all pairwise period ratios, measure proximity to consonant fractions (p/q with $q \le 12$), and assess statistical significance against both uniform and log-uniform null distributions. The log-uniform null is the physically correct comparison, as orbital periods span orders of magnitude and are expected to be approximately scale-free in the absence of resonant interactions [15].

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2. Data and Methods

2.1 Data Source

We queried the NASA Exoplanet Archive (accessed March 2026) for all confirmed exoplanets with measured orbital periods in systems containing two or more planets. This yielded 2,593 planets in 1,038 multi-planet systems. The system multiplicity distribution is: 689 two-planet systems, 216 three-planet, 83 four-planet, 28 five-planet, 12 six-planet, 1 seven-planet (TRAPPIST-1), and 1 eight-planet (Kepler-90).

2.2 Period Ratio Computation

For each multi-planet system, we computed all unique pairwise period ratios. For planets with periods $P_i < P_j$, the ratio is defined as:

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R_{ij} = P_i / P_j
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such that R_{ij} is in $(0, 1]$. This convention ensures all ratios lie in a bounded interval. For a system of N planets, this yields $N(N-1)/2$ unique ratios. Across all 1,038 systems, we obtain 2,344 pairwise ratios.

2.3 Consonance Set: Farey Fractions

We define the consonance set F_Q as the set of all reduced fractions p/q in $(0, 1]$ with denominator $q \le Q$. We adopt $Q = 12$, yielding a consonance set of Farey fractions:

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F_{12} = {1/12, 1/11, 1/10, 1/9, 1/8, 1/7, 1/6, 2/11, 1/5, 2/9, 1/4,
3/11, 2/7, 3/10, 1/3, 4/11, 3/8, 2/5, 5/12, 3/7, 4/9, 5/11, 1/2, 6/11,
5/9, 4/7, 7/12, 3/5, 5/8, 7/11, 2/3, 7/10, 5/7, 8/11, 3/4, 7/9, 4/5,
9/11, 5/6, 6/7, 7/8, 8/9, 9/10, 10/11, 11/12, 1/1}
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Each fraction p/q carries a **consonance rank** $r_c = p + q$. The Consonance Principle predicts that locking strength is inversely proportional to r_c: lower-rank fractions are stronger attractors.

2.4 Proximity Criterion

A period ratio R is classified as **consonant** if there exists a fraction p/q in F_{12} such that:

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|R - p/q| / (p/q) < epsilon
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where epsilon = 0.02 (2% relative deviation). When multiple fractions satisfy this criterion, the ratio is assigned to the nearest one. The 2% threshold is chosen as a conservative window that captures resonant libration amplitudes typical of planetary systems while remaining narrow enough to be statistically discriminating.

2.5 Statistical Null Models

We employ two null models to assess the significance of consonance clustering:

Uniform null. Period ratios are drawn uniformly from (0, 1]. This serves as a naive baseline. Because Farey fractions with q <= 12 are relatively dense in (0, 1], a uniform distribution yields a high expected consonance fraction (~88.3%). This null is physically unrealistic for orbital period ratios.

Log-uniform null. Pairs of orbital periods are drawn independently from a log-uniform distribution over the observed range [P_min, P_max], and the ratio is computed. This is the physically motivated null: if planet periods are scale-free (no preferred absolute timescale), the distribution of log(P) is uniform, and the resulting ratio distribution is concentrated toward extreme values (near 0 and 1). Under this null, the expected consonance fraction is approximately 39%.

For each null, we generate 10,000 Monte Carlo realizations of 2,344 ratios, compute the consonance fraction in each realization, and derive z-scores and empirical p-values.

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3. Results

3.1 Global Consonance Fraction

Of 2,344 pairwise period ratios, 1,967 (83.9%) fall within 2% of a Farey fraction with q <= 12.

Against the log-uniform null (expected fraction ~39%), this yields:

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z = 43.08,  p < 10^{-4}  (0 of 10,000 trials matched or exceeded the observed fraction)
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The observed consonance fraction is more than double the log-uniform expectation, demonstrating that exoplanet period ratios are overwhelmingly concentrated near simple integer ratios at a level far exceeding what scale-free period distributions predict.

Against the uniform null (expected fraction ~88.3%), the observed fraction of 83.9% falls *below* expectation (z = -8.03). This is not evidence against consonance; rather, it reflects the fact that a uniform ratio distribution is an inappropriate null for orbital periods. The deficit relative to uniform expectation is itself informative: it indicates that the ratio distribution has extended tails (from extreme period ratios in widely-spaced systems) that depress the consonance fraction relative to a uniform draw, yet consonance clustering still vastly exceeds the physically relevant log-uniform expectation.

3.2 Most Populated Consonant Ratios

Table 1 lists the twenty most frequently matched consonant fractions.

Table 1. Top 20 consonant fractions by number of period ratios within 2%.

Rank	Fraction p/q	Consonance Rank (p+q)	Hits	Fraction of Total
1	1/2	3	140	5.97%
2	1/3	4	108	4.61%
3	5/11	16	97	4.14%
4	2/3	5	96	4.10%
5	1/4	5	83	3.54%
6	1/5	6	82	3.50%
7	2/9	11	77	3.29%
8	3/10	13	76	3.24%
9	1/12	13	75	3.20%
10	4/11	15	70	2.99%
11	1/6	7	70	2.99%
12	6/11	17	68	2.90%
13	3/11	14	67	2.86%
14	2/5	7	60	2.56%
15	1/7	8	59	2.52%
16	3/7	10	53	2.26%
17	1/9	10	51	2.18%
18	3/8	11	51	2.18%
19	3/5	8	49	2.09%
20	2/11	13	48	2.05%

The four most populated fractions (1/2, 1/3, 2/3, 1/4) are precisely the four lowest-rank Farey fractions in (0, 1], confirming the predicted hierarchy. The 1:2 resonance alone accounts for 140 planet pairs, consistent with its status as the strongest first-order commensurability.

The presence of 5/11 at rank 3 (97 hits) is notable. This high-order fraction (consonance rank 16) appears anomalously populated and likely reflects a geometric selection effect: for systems with moderate period spacing (common among Kepler multis), the ratio P_inner/P_outer frequently falls near 0.45, which lies within 2% of 5/11 = 0.4545. This illustrates the importance of distinguishing true resonant locking from geometric coincidence, a point we return to in Section 4.

3.3 Consonance Rank Distribution

[Figure 1 description] Histogram of consonance rank (r_c = p + q) for the 1,967 consonant-matched ratios. The distribution shows peaks at low ranks: rank 3 (140 hits, exclusively the 1:2 resonance), rank 4 (108 hits, exclusively 1:3), rank 5 (179 hits, combining 2:3 and 1:4), and rank 7 (152 hits, combining 1:6 and 2:5). There is a broad secondary peak at ranks 11-17, reflecting the high-denominator Farey fractions that become available at q = 9-12. The overall envelope is consistent with a 1/r_c dependence modulated by the number of available fractions at each rank (the Farey density increases with rank).

The full rank distribution is summarized in Table 2.

Table 2. Distribution of consonant-matched ratios by consonance rank.

Consonance Rank (p+q)	Number of Fractions at this Rank	Total Hits	Mean Hits per Fraction
3	1	140	140.0
4	1	108	108.0
5	2	179	89.5
6	1	82	82.0
7	2	152	76.0
8	2	108	54.0
9	2	87	43.5
10	2	104	52.0
11	3	187	62.3
12	1	41	41.0
13	4	277	69.3
14	2	94	47.0
15	1	70	70.0
16	2	107	53.5
17	2	133	66.5
18	1	38	38.0
19	1	60	60.0

The mean-hits-per-fraction column reveals the hierarchy most clearly: rank 3 averages 140 hits per fraction, rank 4 averages 108, rank 5 averages 89.5, and rank 6 averages 82. The general trend is decreasing with rank, as predicted, though with scatter at higher ranks due to the geometric selection effects discussed in Section 3.2.

3.4 Validation: Known Resonant Systems

We validate the analysis against six well-characterized resonant multi-planet systems. Table 3 reports the period ratios and deviations from the nearest consonant fraction.

Table 3. Period ratios in known resonant systems.

System	Planet Pair	Period Ratio	Nearest p/q	Deviation
HD 110067	b/c	0.6653	2/3	0.02%
	c/d	0.6667	2/3	0.00%
	d/e	0.7502	3/4	0.03%
	e/f	0.7499	3/4	0.01%
	f/g	0.8001	4/5	0.01%
TRAPPIST-1	all 21 pairs	–	various	all < 0.1%
Kepler-223	b/c	0.7498	3/4	0.03%
	c/d	0.6666	2/3	0.01%
	d/e	0.7502	3/4	0.03%
TOI-178	c/d	0.5714	4/7	0.00%
	d/e	0.6667	2/3	0.00%
	e/f	0.5556	5/9	0.00%
GJ 876	c/b	0.4999	1/2	0.02%
	b/e	0.5003	1/2	0.06%
	e/d	0.2500	1/4	0.00%
Kepler-80	d/e	0.6667	2/3	0.00%
	e/b	0.6668	2/3	0.01%
	b/c	0.6669	2/3	0.02%

All known resonant systems are recovered with deviations well below the 2% threshold, typically < 0.1%. HD 110067 is particularly striking: all five adjacent-pair ratios lock to consonant fractions with deviations < 0.03%, consistent with a pristine resonant chain that has experienced minimal post-formation disruption [3].

For TRAPPIST-1, all 21 unique pairwise ratios (from 7 planets) fall within 0.1% of a Farey fraction. This is the most complete resonant chain known and constitutes a single-system validation of the full Farey hierarchy prediction.

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4. Discussion

4.1 A Universal Mechanism

The central result of this work is that consonance clustering is not a property of select “resonant” systems but a population-level statistical feature of all multi-planet architectures. At $z = 43.08$ against the log-uniform null, the signal is unambiguous: exoplanet period ratios are overwhelmingly attracted to simple integer fractions.

This result extends the classical understanding of MMR in two ways. First, it demonstrates that resonant proximity is the rule, not the exception: 83.9% of all period ratios fall within 2% of a Farey fraction, compared to ~39% expected from a scale-free distribution. Second, it reveals a hierarchy: the lowest-rank fractions (1:2, 1:3, 2:3) are the most populated, with occupancy decreasing with consonance rank, consistent with the prediction that locking strength scales as $1/r_c$.

4.2 The Farey Hierarchy as an Organizing Principle

The Farey sequence has long been recognized in the study of mode-locking in driven oscillators and coupled nonlinear systems [13,14]. Arnold tongues – the parameter regions in which a driven oscillator locks to a rational frequency ratio – are ordered by the Farey sequence, with tongue width proportional to $1/(p+q)$ for the fraction p/q [16]. The Devil’s staircase structure of circle maps produces exactly the hierarchy we observe: low-order rationals have the widest basins of attraction.

Our results demonstrate that planetary orbital dynamics instantiate this mathematical structure at the population level. The consonance rank distribution (Table 2) is a direct observational measurement of the relative widths of Arnold tongues in the gravitational N-body problem, averaged over the full diversity of planetary system architectures.

4.3 Anomalous High-Rank Fractions

The elevated count at 5/11 (97 hits, rank 16) and at several other high-denominator fractions warrants discussion. Two effects contribute. First, a geometric selection effect: the Kepler mission preferentially detected compact multi-planet systems with period ratios in the range 0.3-0.6, where high-denominator Farey fractions are dense. The 2% proximity window around these fractions captures a substantial fraction of ratios in this range. Second, some of these apparent matches may be genuinely locked at high-order resonances, which are expected in systems that migrated rapidly through low-order resonances without capture [9].

Distinguishing true resonant locking from geometric coincidence requires transit timing variation (TTV) analysis, which probes the dynamical libration of the resonant angle. We note that the low-rank fractions (1:2, 1:3, 2:3) are unlikely to be contaminated by this effect, as their prevalence far exceeds any geometric selection.

4.4 Comparison to Previous Work

Fabrycky et al. [12] characterized the period ratio distribution of Kepler multi-planet candidates and identified an excess near 2:1 and 3:2 with an asymmetric peak structure (excess of ratios slightly wide of resonance). Steffen & Hwang [17] quantified this "resonance offset" and attributed it to resonance repulsion via tidal dissipation. Our analysis complements these studies by (a) extending to the full confirmed exoplanet census (not only Kepler), (b) testing all Farey fractions up to q = 12 rather than only first- and second-order MMRs, and (c) quantifying the consonance hierarchy across all ranks.

Notably, the asymmetric peak structure reported in previous work is consistent with our framework: systems slightly wide of resonance are still within the 2% proximity window and are counted as consonant. The offset itself is a signature of the dissipative dynamics that populate resonances – planets migrate into resonance from one side and may be displaced slightly by subsequent perturbations [8].

4.5 Cross-Domain Implications

The Farey hierarchy is not unique to orbital mechanics. Frequency locking at simple integer ratios appears in coupled mechanical oscillators [13], fluid dynamics [18], laser physics [19], and neural oscillations [20]. The mathematical structure is universal: any system of coupled oscillators with dissipation will exhibit mode-locking ordered by the Farey sequence.

This universality suggests a deeper organizing principle. The consonance hierarchy may be understood as a consequence of nonlinear resonance theory applied to any coupled periodic system: the basins of attraction for rational frequency ratios are ordered by the sum of numerator and denominator, with simpler ratios having larger basins. Our exoplanet results provide the largest-scale empirical confirmation of this principle in a gravitational context.

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5. Conclusion

We have demonstrated that 83.9% of 2,344 pairwise orbital period ratios in 1,038 multi-planet systems fall within 2% of a consonant fraction (p/q with q <= 12), a result significant at z = 43.08 against the physically motivated log-uniform null distribution. The most populated ratios follow the Farey hierarchy, with the simplest fractions (1:2, 1:3, 2:3) exhibiting the strongest clustering. All six tested known resonant chains are recovered with sub-0.1% precision.

These results establish consonance locking as a universal statistical property of exoplanet orbital architectures, not merely a feature of exceptional resonant chains. The Farey hierarchy provides a quantitative organizing principle that unifies mean-motion resonance phenomenology across the full diversity of planetary systems.

Predictions. The Consonance Principle makes two testable predictions for future observations: (1) newly discovered multi-planet systems will continue to show ~84% consonance at the 2% level, and the fraction should remain stable as the census grows; (2) systems with higher planet multiplicities will show *higher* consonance fractions, as resonant chains are more dynamically constrained and better preserved in compact architectures.

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